GREAT SCIENTISTS

Galileo Galilei

Italian scientist Galileo Galilei, who is always known by his first name, was born in 1564 near the town of Pisa, Italy. He originally studied to become a doctor, but was much more interested in mathematics. While still a medical student, he noticed a lamp in Pisa Cathedral swaying back and forth. Using his pulse to time the intervals, he worked out that the lamp took the same time to complete each swing, regardless of the length of the arc followed during the swings.

Professor of mathematics

Galileo never became a doctor. He was made Professor of Mathematics at Pisa University at the age of 25 and began to study the physics of motion, as well as engineering. He moved to Padua University and turned his attention to astronomy after building his first telescope in 1609.

Support for Copernicus

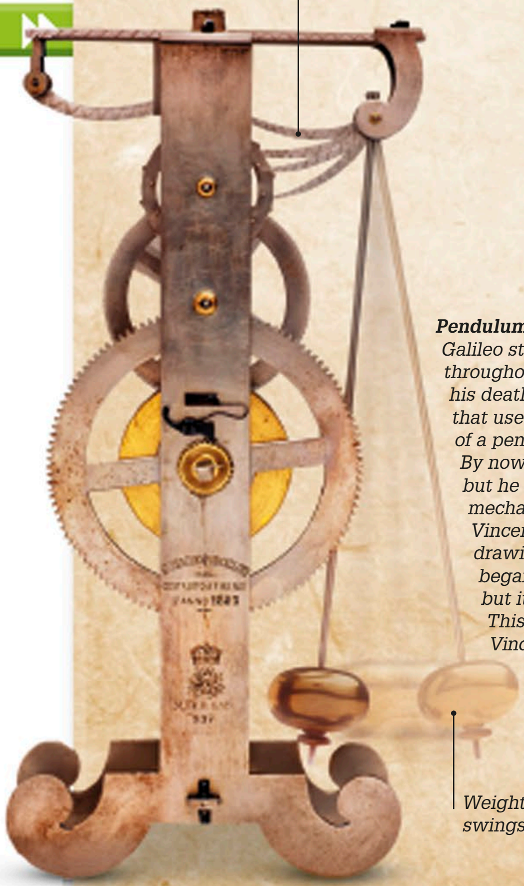
In 1614, Galileo publicly stated his support for the Copernican theory that the planets, including Earth, orbit the Sun. This went against the Church's teaching that Earth was at the center of the Universe and Galileo was told to stop spreading such ideas. He was a devout Catholic and agreed to remain silent, though he was convinced that Copernicus was right.

Final years

Galileo continued with his scientific experiments, but was arrested and put on trial in 1633 for denying the Church's teachings. He spent the rest of his life under house arrest in the village of Arcetri, near Florence. Here he wrote *Dialogues Concerning Two New Sciences*, his last great work on physics, which included his law of falling bodies. He died in 1642.

Surface of the Moon
Galileo was one of the first people to point a telescope at the sky and he made many important observations. He wrote about his discoveries in a book called The Starry Messenger. *Until then, people had thought the Moon was a flat, silvery disc. Galileo's drawings, based on his observations, revealed that it is a sphere with an uneven surface ridged with mountains and craters. The book brought him instant fame.*

> **"It is a beautiful and delightful sight to behold the body of the Moon."**
>
> Galileo Galilei, *The Starry Messenger*, 1610



Lever (pawl) attached to pendulum stops and releases pinwheel with each swing back and forth.

Pendulum clock
Galileo studied pendulums throughout his life. Just before his death, he designed a clock that used the regular sweeps of a pendulum to keep time. By now he was totally blind, but he described the mechanism to his son Vincenzio, who made a drawing of it. Vincenzio began building the clock but it was never completed. This model is based on Vincenzio's drawing.

Weighted pendulum swings to other extreme.

The trial of Galileo
Galileo's trial took place before a crowded Church court in Rome. Charged with heresy and facing a punishment of torture or death, Galileo publicly denied that Earth moves around the Sun. He is said to have muttered defiantly under his breath "And yet, it moves."